HOSPITAL EMERGENCY RESPONSE CO-ORDINATOR

SOURCE CODE

Patient Process

```
#include "hospital_messages.h"

int main(void) {

   printf("[PatientProcess] Connecting to Triage service '%s'...\n",
ATTACH_POINT_TRIAGE);

   fflush(stdout);


   int triage_coid = -1;

   int retries = 10;

   while (retries-- > 0) {

      triage_coid = name_open(ATTACH_POINT_TRIAGE, 0);

      if (triage_coid != -1) break;

      sleep(1);

   }


   if (triage_coid == -1) {

      perror("[PatientProcess] Failed to connect to Triage service");

      fprintf(stderr, "Ensure TriageProcess is running first!\n");

      return EXIT_FAILURE;

   }

   patient_info_t patients[5] = {

      { "P001", "Heart Problem",     9 },

      { "P002", "Breathing Problem", 8 },
```

```
        { "P003", "Leg Injury",       6 },
        { "P004", "Fever",            3 },
        { "P005", "Minor Injury",     2 }
    };

    int num_patients = sizeof(patients) / sizeof(patients[0]);

    for (int i = 0; i < num_patients; i++) {
        hospital_msg_t msg;
        hospital_reply_t rep;
        memset(&msg, 0, sizeof(msg));
        memset(&rep, 0, sizeof(rep));

        msg.msg_type = MSG_TYPE_PATIENT_ADMIT;
        msg.record.patient = patients[i];
I
        printf("[PatientProcess] Submitting patient %s: %s (Severity: %d)\n",
               patients[i].patient_id,
               patients[i].condition,
               patients[i].severity);
        fflush(stdout);

        if (MsgSend(triage_coid, &msg, sizeof(msg), &rep, sizeof(rep)) == -1) {
            perror("[PatientProcess] MsgSend error");
        } else {
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            printf("[PatientProcess] Triage confirmed: %s (Status: %d, Msg: %s)\n",
                patients[i].patient_id, rep.status, rep.message);
            fflush(stdout);
        }

        usleep(300000);
    }
    hospital_msg_t done_msg;
    hospital_reply_t done_rep;
    memset(&done_msg, 0, sizeof(done_msg));
    done_msg.msg_type = MSG_TYPE_BATCH_DONE;
    MsgSend(triage_coid, &done_msg, sizeof(done_msg), &done_rep,
sizeof(done_rep));

    printf("[PatientProcess] All %d patients successfully submitted!\n",
num_patients);
    fflush(stdout);

    name_close(triage_coid);
    return EXIT_SUCCESS;
}
```